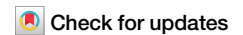


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Controls of root-system overlap on hillslope stability



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More frequent extreme rainfall has heightened the need to understand vegetation's influence on slope stability in steep terrain. Among vegetation-related factors, root-system overlap, the intersection of neighboring root networks, remains understudied despite its potential to influence shallow landslide processes. Here we present laboratory flume experiments testing whether root overlap modulates slope stabilization, using pea plants (*Pisum sativum*) at 50 (V-005), 200 (V-02), 400 (V-04), and 1100 (V-11) stems/m² densities. Landslide initiation was most delayed at V-11, while V-02 and V-005 failed at times comparable to the unvegetated control. V-04 had the smallest landslide area, while lower and higher densities produced larger landslides. Higher plant density increased root reinforcement, yet accelerated soil saturation via preferential flow along overlapping roots. The optimal balance of reinforcement, delaying hydrological triggering, and minimizing landslide size occurred between V-04 and V-11. This study reveals how vegetation density influences slope stability through root-driven geotechnical and hydrological processes.

Vegetation cover has long been considered as one of the key factors influencing shallow landslide occurrence^{1,2}. Various studies have shown that vegetation root systems enhance soil cohesion and alter internal hydrological processes within the soil matrix^{3,4}. The binding force of tree root systems in a forest in Aso, Japan, reduced the risk of slope failure during intense rainfall events³. Reinforcement provided by root systems in forests in the Colorado Front Range helped to mitigate slope instability and debris flow occurrence during an extreme rainfall event in 2013⁴. Moreover, vegetation root systems contribute to the development of preferential flow pathways⁵. Field experiments in coniferous and deciduous forests in China found that finer roots (diameter < 1 mm) enhanced preferential flow⁶, which help to drain water from potentially unstable areas of the slope⁵. In contrast, root systems may also enhance water infiltration during extreme rainfall events, accelerating soil softening and contributing to soil disintegration at the sliding surface, thereby reducing slope stability⁷. Although trade-offs between these stabilizing and destabilizing effects, such as their influence on rainfall thresholds⁸ or failure extent⁹, have been extensively evaluated using various metrics including species and age^{10,11}, vegetation density is often overlooked. Vegetation density affects the availability of below-ground space for individual plants^{12,13}, which in turn shapes root system architecture, encompassing the arrangement and distribution of vertical and lateral roots, and influences how vegetation affects slope stability^{5,14}.

Lateral roots play an important role in stabilizing soil^{15,16}, while vertical roots support soil anchoring⁴, particularly against shallow landslides¹⁷. For example, based on field investigations, landslides occurred below the dense rooting zone due to diminished surficial root reinforcement after clearcutting, highlighting the importance of lateral roots for slope stabilization^{18,19}. The significance of lateral root reinforcement was also observed along basal shear planes in the Oregon Coast Ranges¹⁷. A study in Italian Alps forests showed that lateral roots provided up to four times more reinforcement than vertical roots²⁰. Numerical simulation results showed that lateral roots bind the soil mass helping to resist slope deformation and thus alter the timing and size of landslides²¹.

The interacting effects of neighboring lateral root systems influence reinforcement²². These proximate effects were typically not examined, as most previous studies used only information from individual plants^{22,23}. Numerical simulations suggested that overlap of multiple root systems can help maintain hillslope stability^{22,24}. Based on field observations, mechanically isolated root systems between trees potentially form a weak zone that may initiate slope failure^{22,25}. Because of the limited lateral root system overlap, clustered forest stands with gap distances > 10 m generated a similar factor of safety (FS) of slope stability compared to bare soils, potentially destabilizing the slope²⁵. Similarly, the probability of landslides on a hillslope with a poplar plantation was 70% higher when planting space increased from 10 to 20 m²⁶. Although various field investigations and

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numerical studies demonstrated the importance of tree spacing related to potential root development, the effects of lateral root overlap and resultant slope stability have never been tested using actual vegetation root systems.

Lateral root reinforcement related to the spatial extent of root overlap²² is normally estimated by simply summing individual root contributions on a superimposed area^{16,27}. Hence, such reinforcement tends to increase with stand density as individual plants collectively contribute to the total number of overlapping roots^{28,29}. However, the increase in root reinforcement associated with overlapping roots cannot be assumed to be linear. For instance, a study with Norway spruce showed that tree spacing of 1.5 m had twice the maximum root reinforcement compared to a spacing of 2.5 m²¹. Similarly, decreasing tree spacing from 10 to 5 m resulted in a 34% increase in root reinforcement in an ash forest²⁷. Such nonlinear responses of spacing and root reinforcement relate to plant density because plant community development and succession tend to avoid patches with high densities of neighboring roots due to limited space and competition for nutrients in the soil^{12,13}. As such, simple planar overlapping assessments using information of individual plant root reinforcement may not adequately represent lateral root development and resultant slope stability at the hillslope scale. Furthermore, hydrological processes within soil-root-water interactions can affect the resistance force of root systems^{7,30}; however, these effects are typically not incorporated in evaluations of slope stability at hillslope scales³¹.

Knowledge gaps remain related to how interactions among root growth, root overlap, and internal hydrological processes affect slope stability in natural vegetated conditions with different root densities. To test this hypothesis, we developed a flume experiment to simulate a vegetated hillslope. The flume was 270 cm in length, consisting of two segments that represent landslide initiation (120 cm long, 80 cm wide, and 35° inclination) and deposition (150 cm long, 80 cm wide, and 10° inclination) (Supplementary Fig. 1). We assumed a 1:70 scale for flume dimension and plant growth, corresponding to an 84 m long hillslope and 18 to 21 m tree height (Japanese cedar). While we acknowledge that modeling shallow landslides requires dynamic similarity in soil resistance, hydrology, and failure mechanics, our main goal was to examine how root-system overlap, root structure, and hydrological conditions together influence slope failure across different vegetation densities, rather than to replicate field-scale behavior. We used peas (*Pisum sativum* L.) as a model surrogate for trees because of their rapid growth (~20 cm in 2 weeks) and well-defined root systems (Supplementary Fig. 2c). The tap-like root development and growth of peas consist of a single vertical root supporting lateral roots³²: this root morphology is similar to that of ~73% of woody species³³. Root-to-shoot ratio of peas is 0.2–0.3, similar to root-to-stem ratio for trees (R:S ratio)³⁴; Japanese cedar (*Cryptomeria japonica*) has a R:S ranging from 0.23 to 0.33³⁵. Peas were planted in non-cohesive sand ($D_{50} = 0.23$ mm) at spacings of 3, 5, 7, and 14 cm, corresponding to vegetation densities of 1100 (V-11), 400 (V-04), 200 (V-02), and 50 (V-005) stems/m². We also tested unvegetated conditions (NV) as a control. A constant high-intensity rainfall (90 mm/h) was applied via 30 mist nozzles (0.8 mm in diameter) positioned 2 m above the flume. The time from the onset of rainfall to landslide initiation was measured (hereinafter referred to as timing of landslide). Changes in soil matrix cohesion by root systems were estimated using “root reinforcement” (ΔC) (see **Methods**). Because root mass differed between field conditions and the flume experiment, we introduced scaled-root reinforcement (ΔC^*), normalized by root mass, to compare our findings with previous studies. We also calculated landslide-standardized root reinforcement ($SR = A_s \times \Delta C / W$) where W is landslide weight based on area, depth, and soil density, whereas A_s is area along landslide perimeter. We measured root architectures (i.e., lateral root length, L) and potential overlap characteristics (i.e., overlapping length, OL) of selected individual plants (**Methods**). Soil water content was measured at two depths (3 and 7 cm) along the hillslope transect to assess changes in water storage within the soil matrix. Displacement markers were placed at the soil surface to capture surface deformation and fissure development. All processes, from

rainfall initiation to deformation, landslide occurrence, and mass mobility, were recorded by video from the top and front of the flume (**Methods**).

This study demonstrates that vegetation density, through its influence on lateral root development and overlap, plays a crucial role in modulating slope stability. We found that moderate plant density (V-04) produced the smallest landslide areas and the most consistent failure patterns due to homogeneous root overlap and sufficient reinforcement. In contrast, both sparse (V-005, V-02) and dense (V-11) vegetation treatments showed larger or more variable landslide sizes, either due to insufficient root interaction or excessive hydrological influence in densely packed root zones. These findings indicate a nonlinear relationship between vegetation density and root reinforcement, suggesting the existence of an optimal density that balances mechanical stabilization and hydrological moderation. Our results highlight the importance of incorporating vegetation density and root overlap into slope stability assessments, especially in landscapes dominated by shallow-rooted species or those lacking deep anchorage.

Results and discussion

Vegetation density controls on landslide behavior and root reinforcement

Our flume experiment successfully reproduced actual slope failures with respective morphologies and behavior of material movement based on non-vegetated conditions (NV)^{36,37}. Soil deformation began in the lower slope 0.9 min (± 1 s.d.) after the onset of rainfall (Fig. 1 and Supplementary Video 1). During the primary stage of deformation as rainfall progressed, fissures appeared in the upper hillslope at 1.8 min (± 0.6 s.d.). The slope then entered the secondary stage of deformation, characterized by a steady movement occurring from 1.6 to 2.3 min (Supplementary Fig. 3), possibly associated with soil buoyancy³⁸. From 3.6 to 5.6 min, fissures extended across the slope at maximum deformation rate (dx/dt) of 0.6 mm/s, potentially initiating a tertiary stage of deformation with mean volumetric soil water content ranging from 0.12 to 0.13 m³/m³ (± 0.01 s.d.).

Landslides occurred after 8.2 min of mean value (± 0.8 s.d.) in unvegetated control, with affected areas ranging from 3664 to 6050 cm², corresponding 38 to 63% of flume slope. At the time of landslide occurrence, water contents at the lower slope ranged from 0.3 to 0.31 m³/m³. The saturated soil volume prior to landsliding was 3102 cm³ (± 185 s.d.), representing 51.5% (± 4.3 s.d.) of maximum soil water storage capacity (**Methods**). Landslides were tongue-shaped, similar to rainfall induced landslides observed in the field³⁷. After applying a 70 $\times$ scaling factor, the landslide size measured in the flume corresponded to 1188–3798 m², which falls within the range of more than 90% shallow landslides in the field³⁹. Slip surfaces were located directly above the acrylic base of the flume, at 10 cm soil depth, based on visual observations.

In the various vegetation treatments, vegetation density affected deformation processes, landslide size, and timing of initiation. Soil deformation in vegetated slopes began 1–2.3 min later than in the unvegetated slopes. Fissure development in the upper hillslope occurred at 2.5–9 min, depending on plant density. Among all densities, fissure development occurred earliest in V-05 (4.9 min ± 0.6 s.d.) and latest in V-11 (7.9 min ± 1 s.d.), primarily during the secondary deformation stage. The secondary (after 1–6.3 min onset of rain) and tertiary (after 3–10 min onset of rain) deformation stages in vegetation treatments were also delayed compared to those in NV (Supplementary Fig. 3). Due to high cumulative rainfall at the time fissure development, mean water contents in the tertiary deformation stages of various vegetation treatments (0.08 to 0.21 m³/m³) were higher than in NV (0.13 m³/m³ ± 0.02 s.d.). The maximum deformation rate before landslide initiation was highest in V-04 (23.2 mm/s ± 8.4 s.d.), followed by V-11 (9.3 mm/s ± 7.5 s.d.), V-02 (5.2 mm/s ± 3.1 s.d.), and V-005 (4.5 mm/s ± 2.9 s.d.). During this tertiary stage, numerous fissures (4 to 20 number) developed in the upper hillslope (Supplementary Video 2–5). The timing of landslide initiation was 9.1 min (± 0.67 s.d.) for V-005, 11.6 min (± 2.85 s.d.) for V-02, 14 min (± 0.93 s.d.) for V-04, and 15 min (± 1.45 s.d.) for V-11 (Fig. 2). Landslide sizes ranged from 2426 to 7325 cm²; the smallest was recorded for V-04

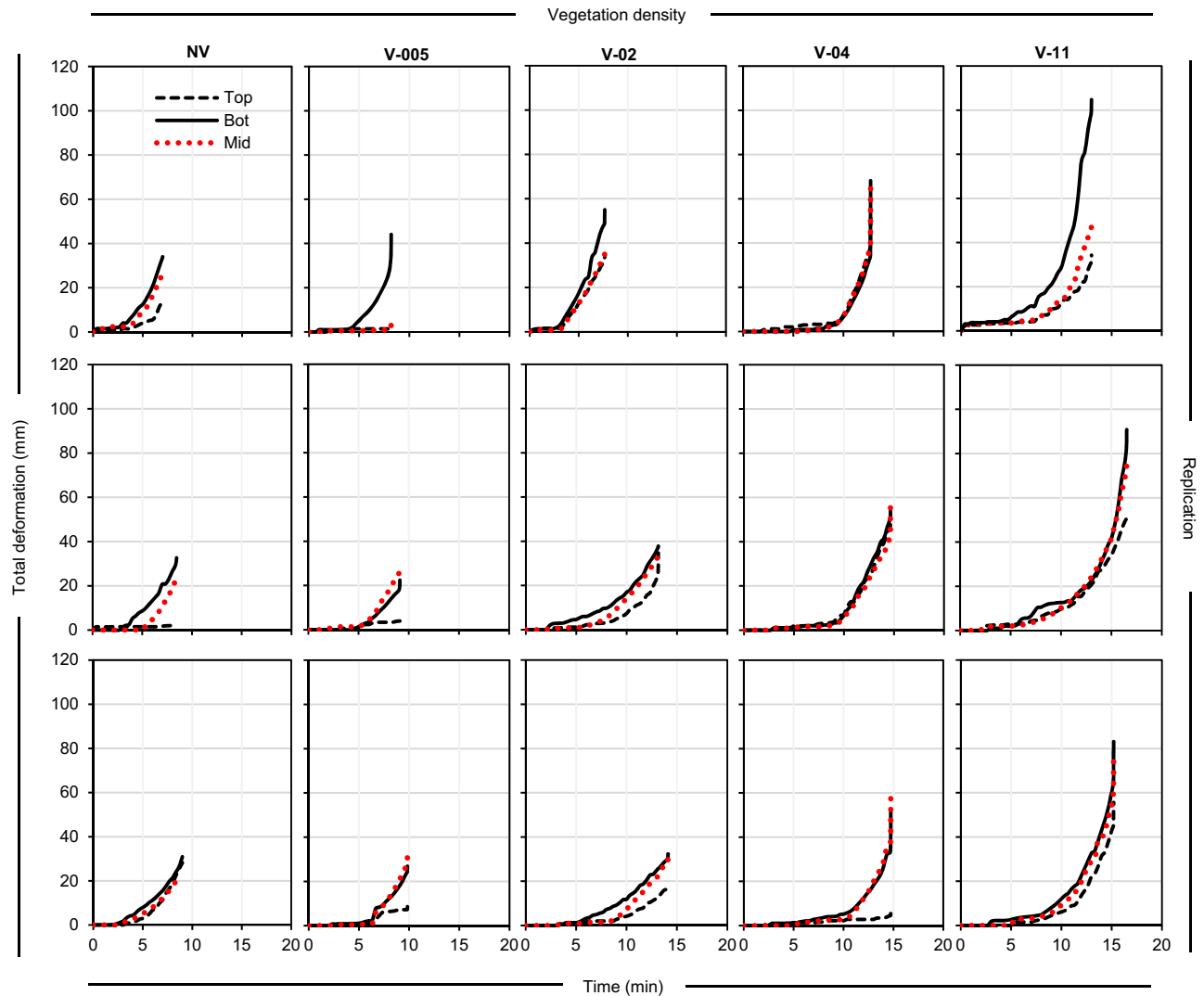


Fig. 1 | Slope deformation (Δx). The black-dashed, black, and red-dotted lines represent the deformation measured at the top, bottom, and middle parts of the slope, respectively. Deformation on NV occurred between 20–120 s, while that on V

was between 61–146 s. NV generated a maximum deformation rate of 0.2–1.2 mm/s, while that on V was 1.8–32.8 mm/s.

(3239 cm² ± 460 s.d.) and the largest for V-11 (4534 cm² ± 2520 s.d.) (Fig. 3). Saturated volume varied from 2904 cm³ in V-005 to 15,676 cm³ in V-011, with a maximum water soil storage ratio ranging from 50–78% (Supplementary Table 1). The sliding interface was an acrylic flume bed situated at 10 cm depth (including NV). Because vertical root anchorage did not occur in either the net or the acrylic flume bed, landslide characteristics (size and timing) among root densities were attributed solely to lateral root development. This setup represented a condition in which trees lacked vertical anchorage, isolating the effects of lateral roots on slope stability.

Soil cohesion provided by root reinforcement (ΔC) increased non-linearly with plant density, while most previous studies focused on the relationship between ΔC and stand age⁴⁰. The five-fold increase in plant density from V-02 (200 stems/m²) to V-11 (1100 stems/m²) resulted in a 15-fold increase in root reinforcement, from 0.16 N/m² (±0.15 s.d.) to 2.5 N/m² (±1.3 s.d.). However, estimated ΔC^* values from our flume experiment were generally lower (3.2–11.1 kN/m² for V-02 to V-011), than previous estimates from Japanese cedar and *Cunninghamia* stands (250–1540 stems/ha; $\Delta C = 13.2$ –19 kN/m²)^{28,29}. On the other hand, landslide-standardized root reinforcement (SR) ranged from 2×10^{-4} to 2×10^{-2} , which is higher than field-based estimates (1.5×10^{-4} – 4.3×10^{-3})⁴¹. These results suggest that the contribution of roots per unit soil mass was greater in our flume experiment than in field conditions. This large root contribution to the landslide mass,

despite the relatively small root cohesion values, may be associated with soil-water-root interactions observed during experiments. Many previous studies estimated root cohesion by summing root tensile strengths under dry conditions and assumed all individual roots within cross-section of the soil mass contributed equally to slope stability (Wu model and Fiber Bundle Model^{22,42}). However, for saturated conditions, roots may lose tensile strength because a lubricating water layer around the roots reduces frictional resistance^{12,43}. Despite the lower ΔC values compared to earlier studies, our approach may yield more realistic root reinforcement estimates, as previous studies may have overestimated this parameter^{27,44}.

Root development and overlap across density gradients

Root architecture development and its hillslope-scale extent among various vegetation densities varied due to spatial constraints and the potential for root-system overlap. High-density lateral roots extended to depths of up to 5 cm, while vertical roots reached nearly 10 cm, down to the acrylic flume bed. Mean root mass, total root length, and number of roots per individual plant decreased with increasing stand density ranging from 1.48 to 1.05×10^{-3} g, 1837 to 1305 mm, and 36 to 27, respectively (Supplementary Table 1). These changes may be attributed to limited space and nutrient availability for individual plant growth^{13,45} due to interspecific competition^{28,46}.

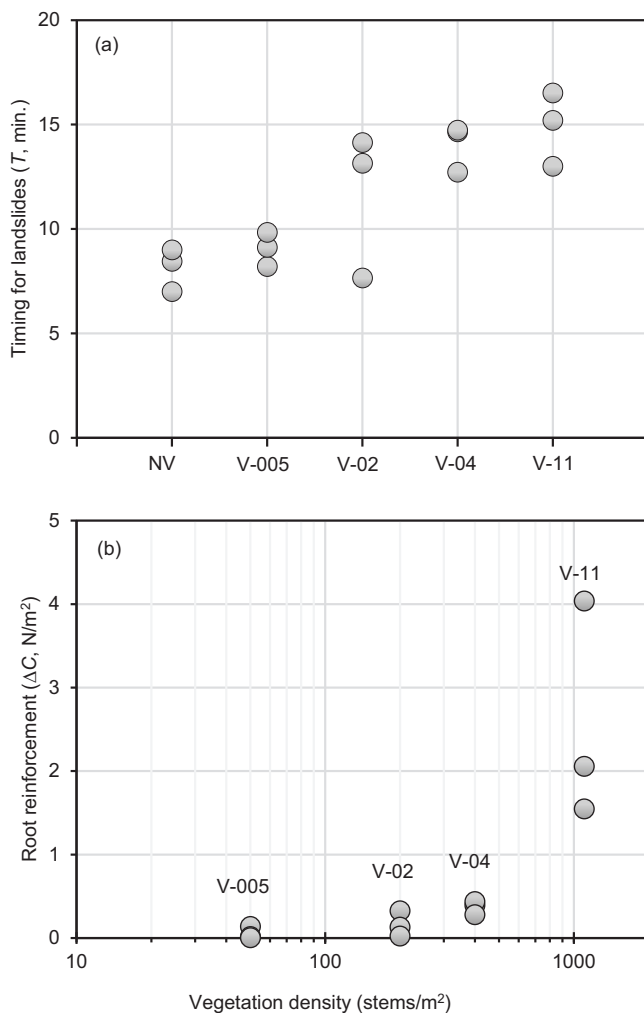


Fig. 2 | Effects of vegetation root systems on slope stability. a Timing for landslide initiation (T). **b** Root reinforcement (ΔC). Each condition has 3 replication data.

At the hillslope scale, mean root mass, total root length, and number of roots increased with plant density, from 0.07 to 1.12 g, 90 to 1392.6×10^3 mm, and 1.8 to 29.4×10^3 , respectively (Supplementary Table 1). Based on plant spacing and root-to-root ratios, overlap percentages, the ratio of the number of overlapping roots (n_o) to the total number of roots (n_t), were estimated to be 0%, 5%, 17%, and 56% for V-005, V-02, V-04, and V-11, respectively. Furthermore, total overlapping, defined by mean overlapping length (OL), total number of overlapping roots (n_o), total overlapping length (OL_t), and overlapping depth (OD) (see Methods), was greatest in V-11, followed by V-04, V-02, and absent in V-005 (Fig. 4). Among all vegetated treatments, both the radial (F_r) and vertical (F_v) correction factors (see Methods) increased with plant density (Supplementary Table 1), indicating that root systems overlapped more homogeneously at high plant densities. Although previous studies^{22,47} employed two-dimensional overlaps using tree spacing and root radius, we estimated overlap length based on root architectures capturing the geometric properties of root systems within the soil matrix.

Controls of root system overlap on root-induced mechanical and hydrological modulation

Changes in the timing of slope deformation and landslide occurrence were associated with root system development and the resultant overlap. As vegetation density increased beyond V-02, maximum deformation increased compared to NV (Fig. 5; Supplementary Table 1). However, the timing of landslide occurrence (also fissure formation) in V-02 showed

greater variability among replicates compared to other treatments (including NV). Based on plant spacing and root radius, the overlap percentages in V-02 were only 0–5%, indicating mechanically isolated or partly overlapped root systems. These results suggest that for sparse vegetation (V-02), soil reinforcement via root overlap may be spatially uneven and only partially effective at the hillslope scale. In contrast, more consistent deformation patterns across replicates in V-04 indicated spatially homogeneous root overlap (Fig. 4a,d) and sufficient reinforcement (Fig. 2b) resulting in landslides that moved as a single block. Although the root system development is more homogeneous at high root densities (V-11), the combination of enhanced overlap, higher root strength (i.e., cohesion based on ΔC), and/or internal hydrological processes may have introduced variability in deformation patterns and mass movement behavior.

Landslide shape and fissure formation on hillslopes provide insight into the presence of weak zones and their relationship to root system development. Weak zones represent potential initiation points of slope failure, particularly where boundaries of landslide masses form^{24,27}. Our results show that in all vegetated treatments, fissures developed predominantly in gaps between plants (Supplementary Fig. 4), suggesting that root-free or sparsely rooted zones act as mechanical weak points. These findings are consistent with those of Roering et al. and Schmidt et al., who observed similar patterns in coniferous and hardwood forests in the Oregon Coast Ranges^{17,48}. The formation of such weak zones is likely influenced by multiple factors including individual root growth characteristics, inter-specific competition, and/or variation of water availability in upper hillslopes, all of which may affect root system development and distribution.

Vegetated treatments with different root densities, overlap characteristics, and root architectures influenced water movement within the soil matrix and the associated timing and extent of saturation zone development. In the unvegetated condition (NV), soil water content increased progressively from shallow to deep layers, indicating downward percolation from the surface. In the deeper soil layer (5–10 cm) of NV, water content responded rapidly ($d\theta/dt = 0.023$ m³/m³ per min), faster than all vegetated treatments ($d\theta/dt = 0.012$ – 0.014 m³/m³ per min). Conversely, vegetated treatments exhibited higher rates of water content change ($d\theta/dt = 0.006$ – 0.008 m³/m³ per min) in the shallow layer compared to NV ($d\theta/dt = 0.0058$ m³/m³ per min). This suggests that root systems and their overlap modulated water distribution across soil layers, from shallow to deep depths and upper to lower hillslopes, potentially via preferential flow pathways⁴⁹. Among the vegetated conditions, V-11, characterized by the greatest root overlap in length and depth, exhibited the highest $d\theta/dt$ values in both soil layers (Supplementary Table 1). As such, the formation of a saturated zone (A_{sat}) at the bottom of the hillslope in V-11 was 1.1–1.4 times faster than in V-04 and V-02 (Fig. 6). This early saturation zone was likely associated with accelerated water movement in both shallow and deep layers. However, NV and V-005 also showed relatively rapid development of saturation zones. The longest extent of the saturated zone at the bottom slope was also observed in V-11. Consequently, landslide initiation timing in V-11 was similar to V-04, despite having 6.7 times greater ΔC , likely due to the enhanced soil mass and reduced soil–root bonding under high saturation conditions. Thus, root system development and the associated hydrological processes influenced both landslide timing and area.

Optimal vegetation density and broader ecological implications

We identified a possible optimal vegetation density for minimizing landslide area (Fig. 7). Landslide areas generally decreased with increasing vegetation density, except in V-11. At high vegetation densities (V-11), roots from adjacent plants grew and intertwined, forming a surface mat^{50,51} (Fig. 5b,c) that reinforced the soil and added tensile strength. During landslide movement, this structure may have contributed to larger mass displacement by holding the soil mass together⁹. Low-density vegetation with minimal root overlap provided insufficient reinforcement, also resulting in large landslides. Other possible reasons for the largest landslide size in V-11 may include a combination of: (1) rapid vertical and lateral water movement in dense root zones near the surface; (2) earlier formation of the saturated zone

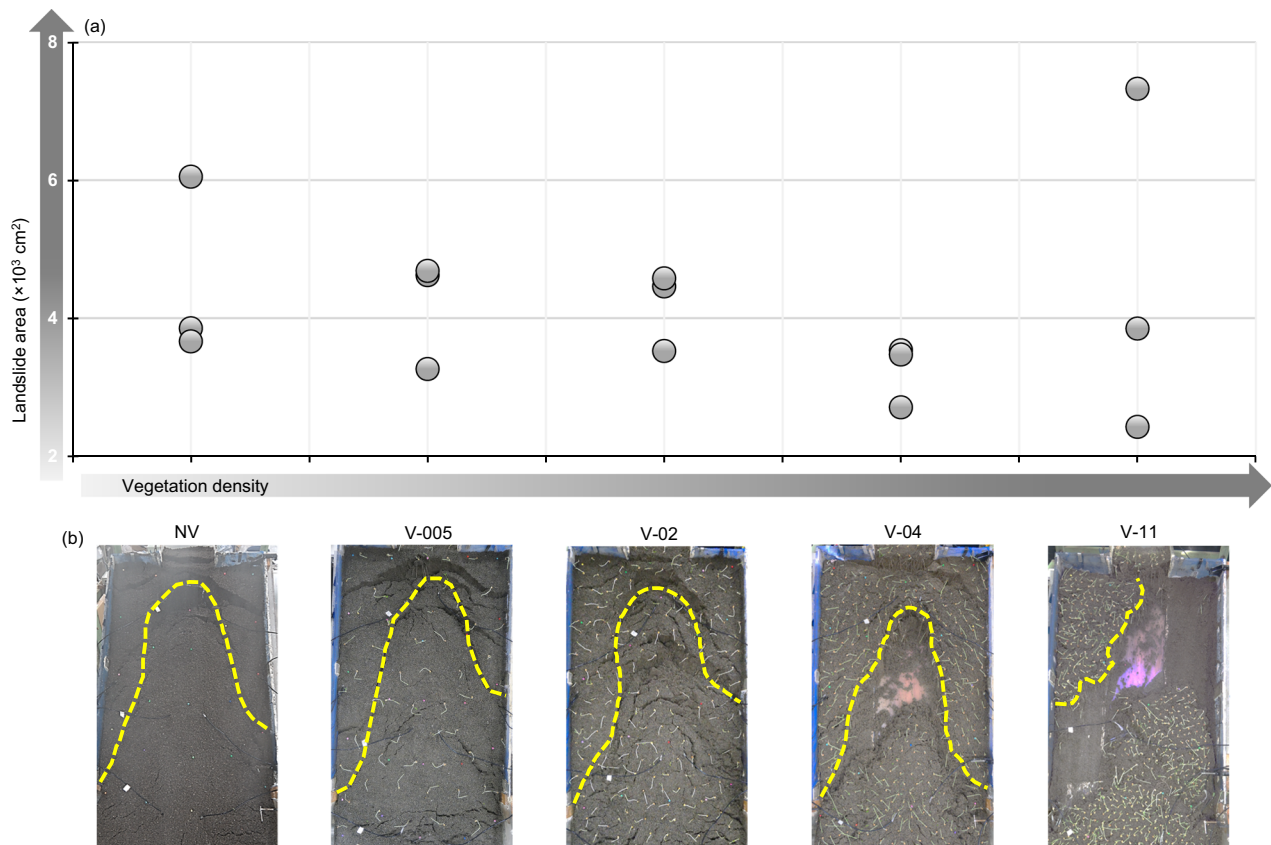


Fig. 3 | Dynamics of slope unstable area. a Landslide area under various vegetation densities. Each condition has 3 replications. **b** Collapse area taken from cameras installed above the flume. The yellow-dashed line indicates slope unstable area.

at the base of the slope; (3) lateral expansion of saturation toward the upper slope; (4) increased cohesion from a dense root network (high ΔC); and (5) binding of the entire soil mass by lateral roots forming a biomat that did not fail during movement (Fig. 5b,c). Therefore, the optimum condition of slope stability in this experiment is likely around V-04. It should be noted that the identified optimum condition reflects a scenario where slope stability was governed solely by lateral root reinforcement, which is particularly relevant for shallow-rooted species or environments with restricted deep anchorage.

Presence of an optimum vegetation condition for minimizing landslide size also provides further insights into the long-standing debate on the relationship between vegetation cover and landslide magnitude^{9,52}. While dense vegetation, particularly on steep slopes with deep soils, can contribute to larger landslides due to the added weight of above- and belowground biomass^{53,54}, vegetation can also play a protective role by reducing landslide size via root reinforcement⁵⁵ and mitigating earthquake induced landslides⁵². In mature forests, substantial root cohesion can decrease landslide size by stabilizing loose or granular soils⁴⁸. While many previous studies examined the vegetation effects on landslide occurrence by types of vegetation, species, and stand ages^{10,11}, our findings highlighted that stand density is a crucial factor in determining whether vegetation decreases or increases landslide area.

Stand density is also important for explaining changes in root strength related to stand age, a widely accepted landslide-root strength model². As trees age increases from 0 to about 10 years, root strength increases in an S-shaped curve, where the initial slow-growth phase reflects the early stages of root system development by young trees with relatively shallow roots¹⁷. As trees mature (~10–20 years), root systems expand both vertically and horizontally, significantly enhancing lateral soil bonding and vertical anchoring, corresponding to the rapid increase in root strength in the middle phase of the S-curve⁵⁶. For example, the extensive root network of a mature mixed forest in China penetrates deeply into the soil, creating a

robust stabilizing effect on a hillslope¹¹. As the tree reaches old age (> 30 years), the rate of root growth slows, although root strength continues to increase, it does so at a diminished rate, forming the plateau phase of the S-curve¹⁰. In contrast, root strength increases nearly linearly or even exponentially with vegetation density, because each additional plant or tree contributes more roots, which cumulatively enhance soil stability through overlapping and intertwined root systems as shown in this study (Fig. 7). However, such collective increases in density are limited in natural ecosystems due to ecological carrying capacity, which constrains the number (or biomass) of plants that can grow and survive a given area⁵⁷. At equilibrium density, plant populations reach a dynamic balance with their environment, where the number of newly established individuals equals the number of individuals dying, a process known as self-thinning⁵⁸. In the contemporary landslide-vegetation interaction model, once a stand matures, the plant community is assumed to reach a stabilizing density at equilibrium, limiting further increases in root strength and cohesion. These ecological constraints illustrate that vegetation's stabilizing role has limits, particularly under conditions of extreme interspecific competition⁵⁸.

Conclusions

Although the optimal condition we found may differ under field conditions depending on vegetation growth rate, species composition, age distribution, and density, our experiment provides insight into how vegetation density influences slope stability through interactions among plant physiological traits, geotechnical properties, and hydrological dynamics. Our findings also highlight that effective management of vegetation density, whether in natural ecosystems, agriculture, or forestry, requires a deep understanding of these complex interactions to ensure long-term sustainability and resilience. This understanding, in turn, can inform the assessment and management of stand density in relation to age structure and competition, helping ensure that ecosystems remain healthy, productive, and diverse over time.

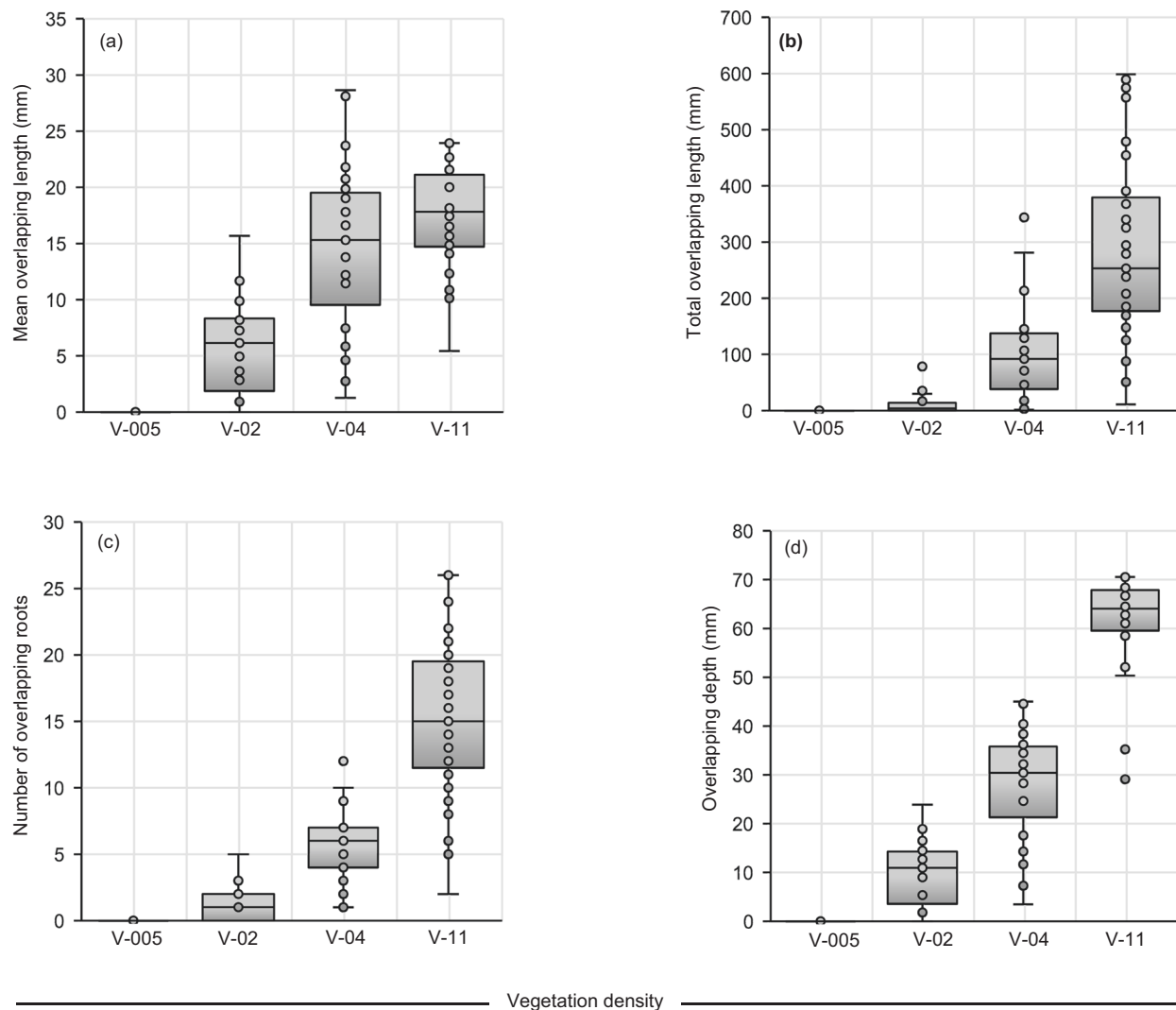


Fig. 4 | Root overlap parameters on the slope. a Mean overlapping length. **b** Total overlapping length. **c** Number of overlapping roots. **d** Overlapping depth. Boxes cover the interquartile range (IQR). Straight line inside boxes indicates median. Round markers are the data. Each condition has 27 data.

Understanding the relationship between plant density and slope stability is essential for designing effective and sustainable Nature-Based Solutions (NbS) that align with natural ecosystem dynamics, ensuring long-term resilience and optimal resource use.

Methods

Experimental setup

This flume experiment focused on simulating rainfall-induced shallow, rapid landslides. The flume consisted of two segments that represent landslide initiation (120 cm long and 80 cm width, 35° inclination) and deposition (150 cm long and 80 cm width, 10° inclination). In the slope segment, a net (5×5 mm mesh) was installed on the flume bed with a 30×80 cm² open area at its center to vary the basal resistance along the flume, thereby establishing the boundary lines for potential landslides. A similar setup with clearance on the center of the slope was also used in a numerical simulation to investigate the control of tree roots on shallow mass movement²¹. To simulate rainfall, thirty UXCELL mist nozzles with 0.8 mm diameter and 80–145 ml/min outflow discharge were installed 2 m above the slope at 30 cm intervals (Supplementary Fig. 1d,e). A flowmeter with a 30 l/min discharge capacity was installed at the inlet to measure the inflow discharge. For each 0.5 l/min of inflow measured by the flowmeter, rainfall intensity was measured by placing a tipping bucket (0.2 mm/tip of resolution) under the nozzles, repeated three times for each value of discharge. To

control rainfall using inflow discharge, a calibration between discharge and rainfall intensity was developed using simple linear regression. For instance, 90 mm/h of rainfall intensity is required for a 1.73 l/min of inflow discharge. To simulate shallow soil and growing medium for plants, non-cohesive sand ($D_{50} = 0.23$ mm; bulk density = 1.4 g/cm³) was placed on the slope by using a small shovel (1 kg capacity) without compaction until it reached a depth of 10 cm. Six EC-5 decagon FDR sensors were installed at 3 and 7 cm below the soil surface to measure soil-water contents (Supplementary Fig. 1c). These sensors were calibrated using a linear calibration equation constructed from the actual water contents in laboratory tests and measured water content values from the sensors. Three video cameras were installed above and in front of the slope to capture soil movement (Supplementary Fig. 1a). Displacement was also measured using twenty-five markers (1 cm diameter colored plastic beads) installed on the soil surface. A tipping bucket (100 ml/tip resolution) was installed at the end of the flume to measure outflow.

Preparation of vegetation

We used dun peas (*Pisum sativum* L.) to simulate root systems in the hillslope. Pea seeds were initially soaked in water at 20 °C temperature for 24 h. The seeds were then rinsed and wrapped in a wet towel for 2 days until sprouting. Only sprouts with straight main roots (2–3 cm in length) were planted on the slope following a hexagonal pattern to maximize root overlap (Supplementary Fig. 2a,b). The sprouts were planted on the slope

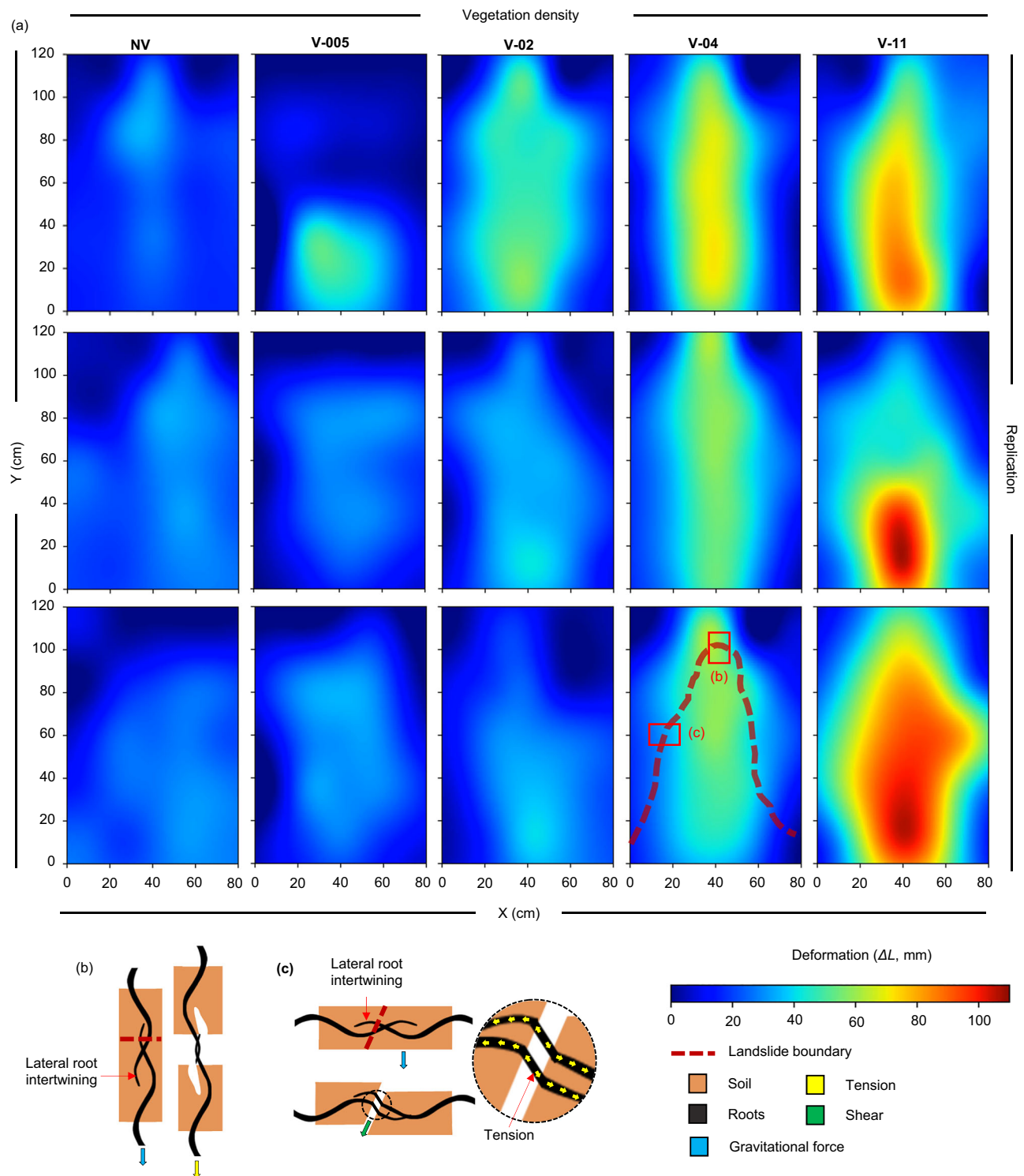


Fig. 5 | Deformation during landslide occurrence. **a** Spatial distribution of deformation values. X represents the flume width, and Y represents the flume length, where $Y = 0$ and $Y = 120$ correspond to the lower and upper parts of the slope, respectively. The X-Y plane represents the top view of the slope. The color inside the box represents deformation values. The maximum deformation was observed in

V-11 ($99 \text{ mm} \pm 9.6 \text{ s.d.}$), followed by V-04 ($60.3 \text{ mm} \pm 10.7 \text{ s.d.}$), V-02 ($45.6 \text{ mm} \pm 9.5 \text{ s.d.}$), V-005 ($36.6 \text{ mm} \pm 8.1 \text{ s.d.}$), and NV ($32.6 \text{ mm} \pm 1.3 \text{ s.d.}$). Differences in deformation values along the Y-axis indicate tensile or compressive deformation, while differences along the X-axis indicate shear deformation. **b** Illustration of tensile deformation. **c** Illustration of shear deformation.

with 3, 5, 7, and 14 cm intervals to simulate root systems of 1100 (V-11), 400 (V-04), 200 (V-02), and 50 (V-005) stems/ m^2 , corresponding to 2200, 800, 400, and 100 stems/ha of the actual forest, respectively ($70 \times$ scaling). About 0.8–1.0 liters of water was applied once every two days, and no fertilizer was added. Since the sprouts were grown indoors,

artificial blue and red lights were used to simulate daylight in summer with an exposure time of 12–14 h/day for photosynthesis. The room temperature was maintained at 27°C using air conditioners. The peas reached 25–30 cm in height after 2 weeks before the experiment, which is comparable to 17.5–21 m tall trees using $70 \times$ scaling.

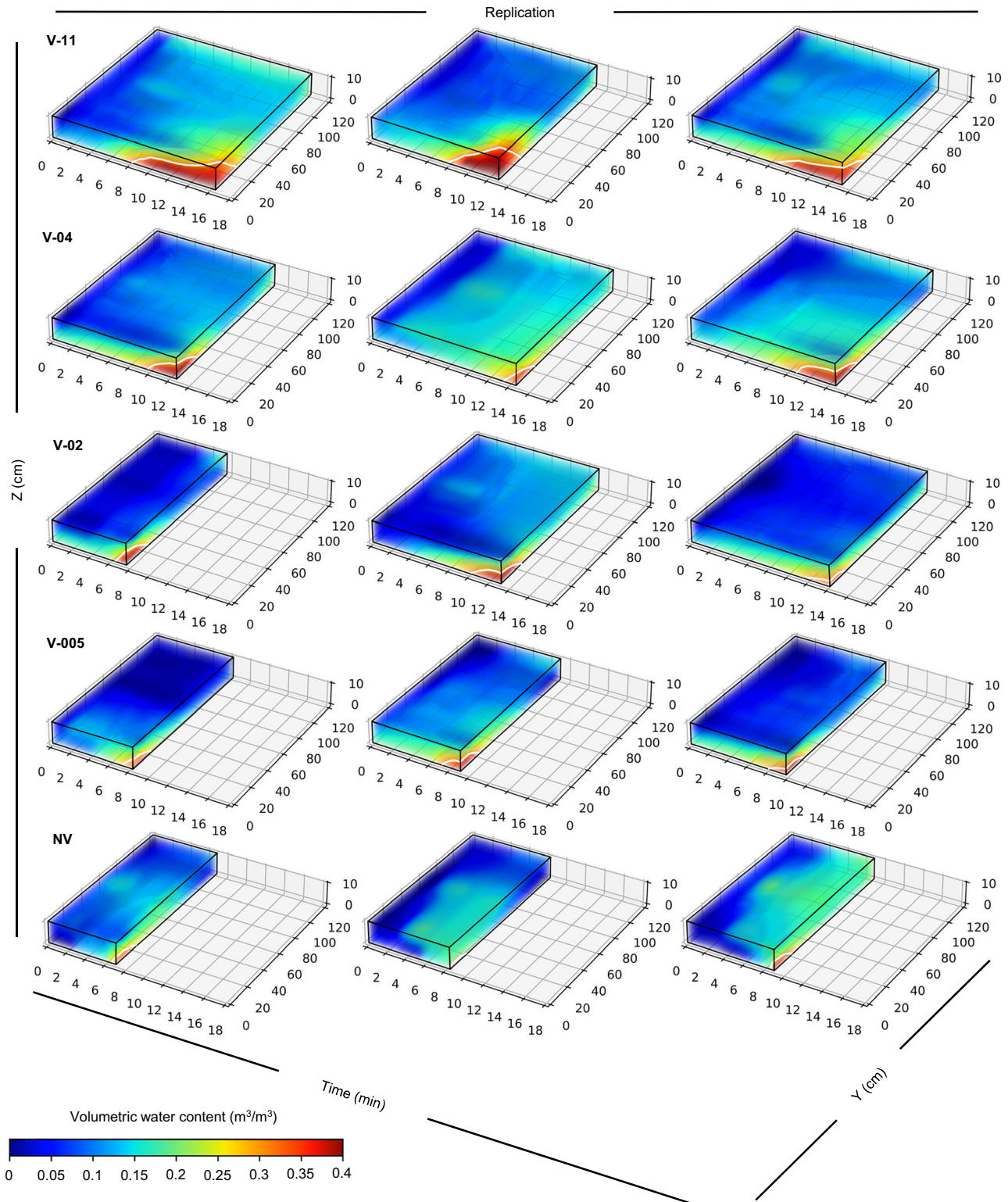


Fig. 6 | Spatiotemporal distribution of water content. Y is flume length, where $Y = 0$ and $Y = 120$ are the lower and upper parts of the slope, respectively. Z is soil depth, where $Z = 0$ and $Z = 10$ are flume bed and soil surface, respectively. X-Z plan shows the longitudinal distribution of water content. Time axis represents the time from the onset of applied rainfall ($t = 0$) to landslide initiation (end of the box). Areas under the white line are saturated zone (A_{sat}).

A_{sat} during landslides was the largest for V-11 ($164.4 \text{ cm}^2 \pm 44.2 \text{ s.d.}$), followed by V-04, ($74 \text{ cm}^2 \pm 6.7 \text{ s.d.}$), V-02 ($44.6 \text{ cm}^2 \pm 10.2 \text{ s.d.}$), V-005 ($31.7 \text{ cm}^2 \pm 9.7 \text{ s.d.}$), and NV ($23.1 \text{ mm}^2 \pm 10.1 \text{ s.d.}$). These saturated areas were formed at 8.2 min ($\pm 0.2 \text{ s.d.}$) for V-11, 11.2 min ($\pm 1.9 \text{ s.d.}$) for V-04, 9.4 min ($\pm 2.9 \text{ s.d.}$) for V-02, 7.3 min ($\pm 0.6 \text{ s.d.}$) for V-005, and 7.3 min ($\pm 1.2 \text{ s.d.}$) for NV.

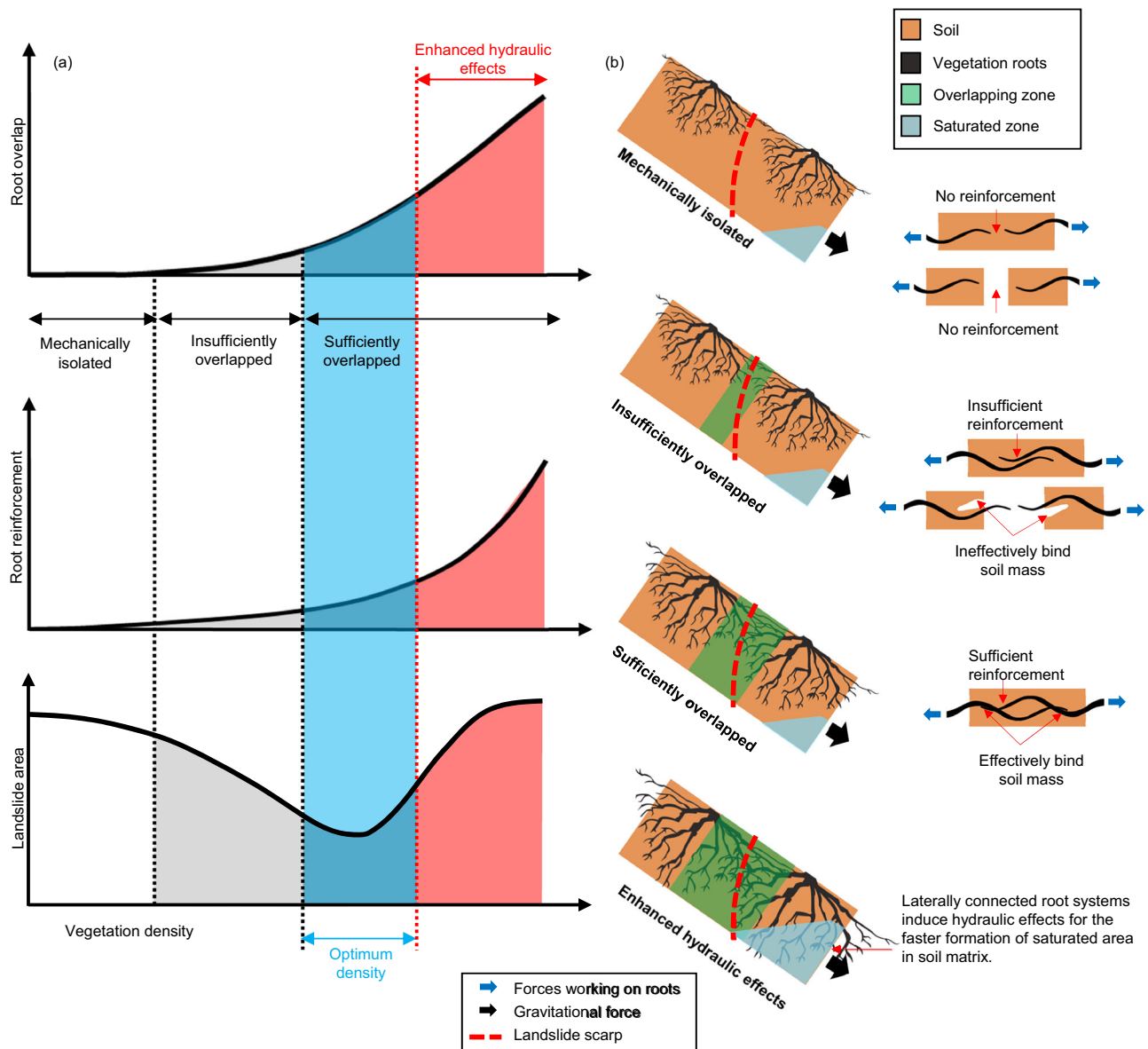


Fig. 7 | Controls of root systems overlap on shallow landslides. **a** Schematic diagram of the effect of root systems overlap on root reinforcement (ΔC) and landslide area (A). **b** Possible root failure mechanisms under various overlapping conditions. Vegetation root systems are mechanically isolated when roots do not overlap (e.g., V-005), resulting in minimal influence on ΔC . At low vegetation densities, slightly overlapping root systems (e.g., V-02) provide insufficient reinforcement to the soil mass, leading to a slight increase in ΔC and a reduction in A . In contrast, sufficiently overlapping root systems (e.g., V-04) offer adequate reinforcement, significantly

improving ΔC and effectively reducing A . At higher vegetation densities, highly overlapping root systems (e.g., V-11) reinforce the soil mass more effectively, substantially increasing ΔC . However, this condition can induce hydraulic effects, accelerating the formation of saturated areas within the soil matrix and ultimately increasing A , despite the high ΔC value. The optimum condition in this illustration represents a case where slope stability depended entirely on lateral root reinforcement, especially relevant for shallow-rooted species or settings with limited deep anchorage.

Experimental procedures

Prior to water application, the aboveground biomass of peas was cut to a height of 3–5 cm to minimize rainfall interception and avoid local concentration of raindrop impact via vegetation stems (reducing local concentration of pore-water pressure accretion from stemflow and throughfall). A constant rainfall of 90 mm/h was then applied from the nozzles by controlling the inflow discharge to 1.73 l/min. The time and location for fissure formation were recorded based on visual observation, then later confirmed by video. The timing of landslide initiation was measured relative to the onset of rainfall application. When landslides occurred, we measured root system properties, landslide size, deformation, water content, and root reinforcement. The same procedure was repeated 3 times for each condition. The datasets for all experiments are provided in Supplementary Data 1.

Measurement of properties of root systems

Root system properties per individual plant (i.e., root size, root length, number of roots, and root mass) were measured from nine intact root system samples collected from upper ($n = 3$), middle ($n = 3$), and lower ($n = 3$) parts of the slope by washing the soil-root samples with water after each experiment. We measured root size (diameter) and length by using a digital caliper with 0.01 mm of resolution for each individual root within the root system. To account for differences in root diameter along their lengths, measurements were taken at the midpoint of the root length. Total root length was then estimated by summing the length of all individual lateral roots. The number of roots corresponds to roots > 1 cm in length. To account for the vertical root distribution, the depth of each lateral root (z : mm) was measured (Supplementary Fig. 2d). Root mass was measured by

oven-drying root samples at 75 °C for 48 h and weighting using a digital scale with 0.0001 g of resolution.

Root system properties for the entire slope (i.e., total root length, total root mass, and total number of roots) were estimated by averaging values from individual vegetation levels multiplied by the total number of plants on the slope. Since root overlap is the key factor for root reinforcement at the hillslope level²², we also estimated root overlap parameters (i.e., number of overlapping roots, total overlapping length, and mean overlapping length). Due to the effect of gravity, the growth of the lateral roots was assumed to follow an elliptical path, extending vertically and laterally with soil depth (z : mm) and distance between vegetation (I : mm), respectively (Supplementary Fig. 2f). Thus, the length function of each individual lateral root (L_i : mm) can be expressed as follows, where x_i represents the horizontal projected length of roots (mm), which can be obtained through a solver function in MS EXCEL.

$$L_i(x_i) = -\frac{z^2 x_i \sqrt{I^2 - x_i^2}}{2I^2} + \frac{1}{2} z^2 \tan^{-1} \left(\frac{x_i}{\sqrt{I^2 - x_i^2}} \right) + x_i \quad (1)$$

The lateral overlapping length (OL : mm) was then estimated by using a simple trigonometric equation based on the x -axis projection of individual root length (x_i : mm), lateral root inclination (α : °), and distance between plants (I : mm) (Supplementary Fig. 2d). Lateral root inclination was estimated based on the shortest distance from the base to the tip of the root (Supplementary Fig. 2d), measured by removing soil at the stable zone of the slope after each experiment. Individual roots overlap when OL_i (mm) > 0.

$$OL_i = 2x_i - I \quad (2)$$

To account for the radial distribution of individual roots on overlapping calculations, we developed a radial distribution correction factor (F_r) based on the ratio of the overlapping area (Supplementary Fig. 2b) and root radius area. Here, x_{max} represents maximum lateral projection of root length or root radius (mm), whereas OL_{max} is the maximum overlapping length (mm).

$$F_r = \frac{6 \left[\frac{1}{\pi} \sin^{-1} \left(\frac{x_{max} - \frac{1}{2} OL_{max}}{x_{max}} \right) \pi x_{max}^2 - \sqrt{\frac{1}{4} OL_{max}^2 - x_{max} OL_{max} \left(x_{max} - \frac{1}{2} OL_{max} \right)} \right]}{\pi x_{max}^2} \quad (3)$$

As individual roots were also vertically distributed, we calculated the overlapping depth (OD_i : mm) representing the depth at which the individual roots still overlapped (Supplementary Fig. 2d). Mean overlapping depth (OD : mm) was then estimated by averaging OD_i .

$$OD_i = \frac{z}{I} \sqrt{I^2 - x_i^2} \quad (4)$$

Hence, not all roots in the vertical direction overlap. As such, we developed a vertical correction factor (F_v) based on the maximum overlapping depth (OD_{max} : mm) and soil thickness (z : mm) (Supplementary Fig. 2d).

$$F_v = \frac{OD_{max}}{z} \quad (5)$$

Total overlapping length at individual vegetation levels (OL_i : mm) was then estimated by summing OL_i , corrected by F_r and F_v . Similarly, the mean lateral overlapping length at the individual vegetation level (OL : mm) was

estimated by averaging OL_i , corrected by F_r and F_v (Eq. 6).

$$OL = \frac{\sum_{i=1}^n OL_i}{n_o} F_v F_r \quad (6)$$

where n_o represents total number of overlapping roots within a root system.

Landslide size measurement

Landslide size (area, A) was estimated by using a rectangular multiplication of mean landslide width (B_i) and mean landslide length (D_i) measured after landslide occurrences (Eq. 7).

$$A = \frac{\sum_{i=1}^n B_i \times \sum_{i=1}^n D_i}{n^2} \quad (7)$$

B_i and D_i were measured with a metric tape (1 mm resolution) at five equally spaced widths and lengths of landslides to ensure an accurate estimation of A .

Deformation measurement

We estimated slope deformation based on the movement of 25 displacement markers installed on the soil surface based on video monitoring (Supplementary Fig. 1c). Deformation for each marker was detected from interval cameras and measured by using AutoCAD 2021 Student Version at 20 s intervals (Eq. 8).

$$\Delta x = S_f \times (x_i - x_0) \quad (8)$$

where, Δx is deformation (mm), x_i and x_0 are marker positions at $t = i$ and 0 s (mm), measured from the reference point, respectively. S_f is the scaling factor (ratio between the actual slope length and measured slope length from AutoCAD 2021). We used the same reference point for all deformation measurements. We then created the spatial distribution of maximum deformation values on an x - y axis (plan view) by interpolating measured deformation values (see Supplementary Data 2) using `scipy.interpolate()` module in Python with a 1 cm × 1 cm resolution.

Water content measurement

The spatial distribution of water content values along the y - z axis (longitudinal direction) was estimated by interpolating water content values obtained from FDR sensors. Interpolation was done by using the `scipy.interpolate()` module in Python with a 1 cm × 1 cm resolution. The same interpolation technique was used for different water content values at different times (from rainfall application to landslide generation at 1 s intervals, see Supplementary Data 3) to create temporal patterns of water content (x - z axis). When locations on the y - z plane initially reached saturation (water content > 0.3 m³/m³), we recorded the timing and considered it as the timing for saturated area formation. Saturated area was then estimated based on the total area with water content > 0.3 m³/m³ on the y - z plane.

The water storage ratio (R) was measured from water storage in soils and total inflow volume at t_i seconds. Water storage (ΔS : m³) was estimated by multiplying the average water content from FDR sensors and total soil volume, assuming no volumetric soil change. Total inflow volume was estimated from inflow discharge and time from rainfall application. $\bar{\theta}_i$ and $\bar{\theta}_0$ represent the mean volumetric water contents at $t = i$ and 0 s (m³/m³), respectively. B , D , and h are slope width, length, and soil thickness (m), respectively. Q_i is inflow discharge (m³/s), whereas t_i is time measured from rainfall application (s) (Eq. 9).

$$R = \frac{(\bar{\theta}_i - \bar{\theta}_0) BDh}{\sum_{i=0}^n Q_i \times t_i} \quad (9)$$

Root reinforcement estimation

Root reinforcement can be expressed as an additional cohesion (ΔC) in the Coulomb failure criterion^{59,60}. In general, the strength of the soil-root composite (S) can be expressed as follows (Eq. 10).

$$S = C' + \Delta C + (\sigma - u) \tan \phi \quad (10)$$

where C' is soil effective cohesion, σ is normal stress due to overburden pressure, u is pore-water pressure, and ϕ is internal friction angle of soils. Back-calculation is widely used to estimate ΔC at the hillslope level, particularly if geotechnical and hydrological data are available². By using 3-dimensional force equilibrium at critical conditions (Factor of Safety = resisting/driving forces = 1), back-calculated ΔC (N/m²) can be expressed as follows (Eq. 11).

$$\Delta C = \frac{\sum_{i=1}^n [W_i \sin \alpha - (W_i - u_i) \cos \alpha \tan \phi - (C'_b B_i D_i + C'_l D_i h_i)]}{2 \sum_{i=1}^n D_i h_i} \quad (11)$$

where, W represents landslide weight (N), u is pore-water pressure (N/m²), and C'_b and C'_l are effective soil cohesion at the slip surface and landslide perimeter (assumed zero for non-cohesive materials⁶¹), respectively. B , D , and h are landslide width, length, and thickness (m), respectively, and α represents slope inclination (35°). Since the landslide body is analyzed as a finite number of slices⁶², i represents the slice number and n denotes total number of slices. The resultant of inter-slice forces is assumed zero since they negate each other⁶³. Pore-water pressure (u) was assumed to affect only the saturated zone with values equal to the height of the saturated zone multiplied by water density (1 g/cm³) and gravitational acceleration (981 cm/s²). The saturated zone was estimated based on area with water content > 0.3 m³/m³, predicted by interpolating values from FDR sensors. Mean thickness of saturated zone is estimated as area/length of the saturated zone. Since internal friction angle (ϕ : °) data was not available, this parameter was back-calculated for non-vegetated conditions (Eq. 12); ϕ was estimated as 35.5–35.7° (± 0.1 s.d.), typical of sandy material.

$$\phi = \tan^{-1} \left[\frac{W_i \sin \alpha}{(W_i - u_i) \cos \alpha} \right] \quad (12)$$

Since root reinforcement depends strongly on the distribution of root mass in soils^{28,29}, we scaled up our estimated ΔC by using the root mass. Prior investigation showed that root mass is a good indicator for explaining soil reinforcement provided by plant roots⁶⁴. Hence, we construct a root mass-root reinforcement conversion graph by using linear regression of total root mass per square meter and back-calculated ΔC obtained from this study. This conversion equation is expressed by Eq. 13, where ΔC is scaled-up root reinforcement in kN/m² and m is root mass per square meter.

$$\Delta C = 1.8m \quad (13)$$

Data availability

All data supporting the results of this study are publicly accessible through Figshare at <https://doi.org/10.6084/m9.figshare.29579462>.

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Author contributions

RN designed and performed the experiment, analyzed data, produced figures, and wrote the paper as the major contributor. TG organized and guided the study, designed the experiment, developed analytical method, and edited the paper. RCS developed analytical method and edited the paper. NI designed the experiment and developed analytical method. NO designed the experiment.

Competing interests

The authors declare no competing interests.

Additional information

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